



Chandrayaan-2 TMC2 PDS4 Data Products User Guide



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Space Applications Centre Ahmedabad

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Chandrayaan-2 TMC2 PDS4 Data Products User Guide

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1. Introduction

1.1 Purpose and Scope

The purpose of this user guide document is to bring about details about the Chandrayaan-2 PDS4 Data Products from the user's perspective. It lists out the number of data products, types and formats available for the users to work with PDS4 data products.

1.2 Audience

This report is meant for the staff of archiving authority (ISSDC for ISRO), data processing team, PI team, and users from scientific community.

1.3 Definitions, Acronyms & Abbreviations

Table 1 brings out the list of definitions, acronyms & abbreviations used in this report.

Table 1 Definitions, Acronym & Abbreviation

Acronym	Abbreviations/Description
PDS	Planetary Data System
ISRO	Indian Space Research Organization
ISSDC	Indian Space Data Centre
ISDA	ISRO Science Data archive
SAC	Space Applications Centre
DP	Data Processing
PI	Principal Investigator
DEM	Digital Elevation Model
ins	Instrument id

1.4 Applicable Documents

1. Chandrayaan-2 Science data management and Archive plan (Chandrayaan-2/DP /SAC/SIPG/HRDPD/TR-06/July 2018)
2. Chandrayaan-2 SAC DP – ISSDC Interface for Archival (Chandrayaan-2/DP /SAC/SIPG/HRDPD/TR-13/Dec. 2018)

2. Chandrayaan-2 PDS4 Data Archive

Chandrayaan-2, India's second mission to the Moon was launched by GSLV-MK-III in July 2019. Mission carries 13 scientific payloads on Orbiter. Orbiter is presently orbiting moon under 100km x 100 km circular orbit.

The data collected from the instruments from the orbiter will be received, processed and archived in Indian Space Science Data Center (ISSDC) for dissemination and use by scientific community in India and abroad.

The data from Chandrayaan-2 instruments should be archived in such a way that they are easily accessible by Internet through simple interfaces being used elsewhere for similar kind of missions.

PDS4 is the de facto international standard for long term archival of planetary science data. For all the scientific payloads, PDS4 data products types [1] were identified and shown in the below **Table 2**.

Table 2 Chandrayaan-2 PDS4 Data Products Types

Sr. No.	Instruments	PDS4 Data Products Types		
		Raw	Calibrated	Derived
1	TMC2	Yes	Yes	Yes
2	IIRS	Yes	Yes	Yes
3	OHRC	Yes	Yes	No
4	SAR	Yes	Yes	No
5	CLASS	Yes	No	No
6	XSM	Yes	Yes	No
7	CHACE-2	Yes	No	No

Each of the PDS4 data products types were mapped to the ISRO Levels and CODMAC Level as shown in **Table 3**.

Table 3 Chandrayaan-2 PDS4 CODMAC Mapping

PDS4 Level	PDS4 Level Description	ISRO Level	CODMAC Level
Raw	Original data from an instrument after initial pre-processing like decompression, reformatting, packetization etc.. the archived data are in a PDS approved archive format.	0	2
Calibrated	Data converted to physical units, which makes values independent of the instrument (Radiometrically corrected & seleno-tagged)	1	3
Derived	Results that have been distilled from one or more calibrated data products (for example, maps, gravity or magnetic fields, or ring particle size distributions). Supplementary data, such as calibration tables or tables of viewing geometry, used to interpret observational data should also be classified as 'derived' data if not easily matched to one of the other three categories (higher level products like ortho, DEM)	2	4+

3. Archive Structure

Archive structure is designed at instrument level and data product level

3.1 Instruments Collection and Data Products

Next level instrument wise collection is defined based on payload name for example for tmc2 payload, tmc_collection is defined. Under each instrument wise data collection, data is organized under each directory defined by PDS4. Data is organized based on the levels of data processing and data products definitions. All the data is planned to store as Year and month wise. **Figure 1** shows detail instruments wise collection structure.

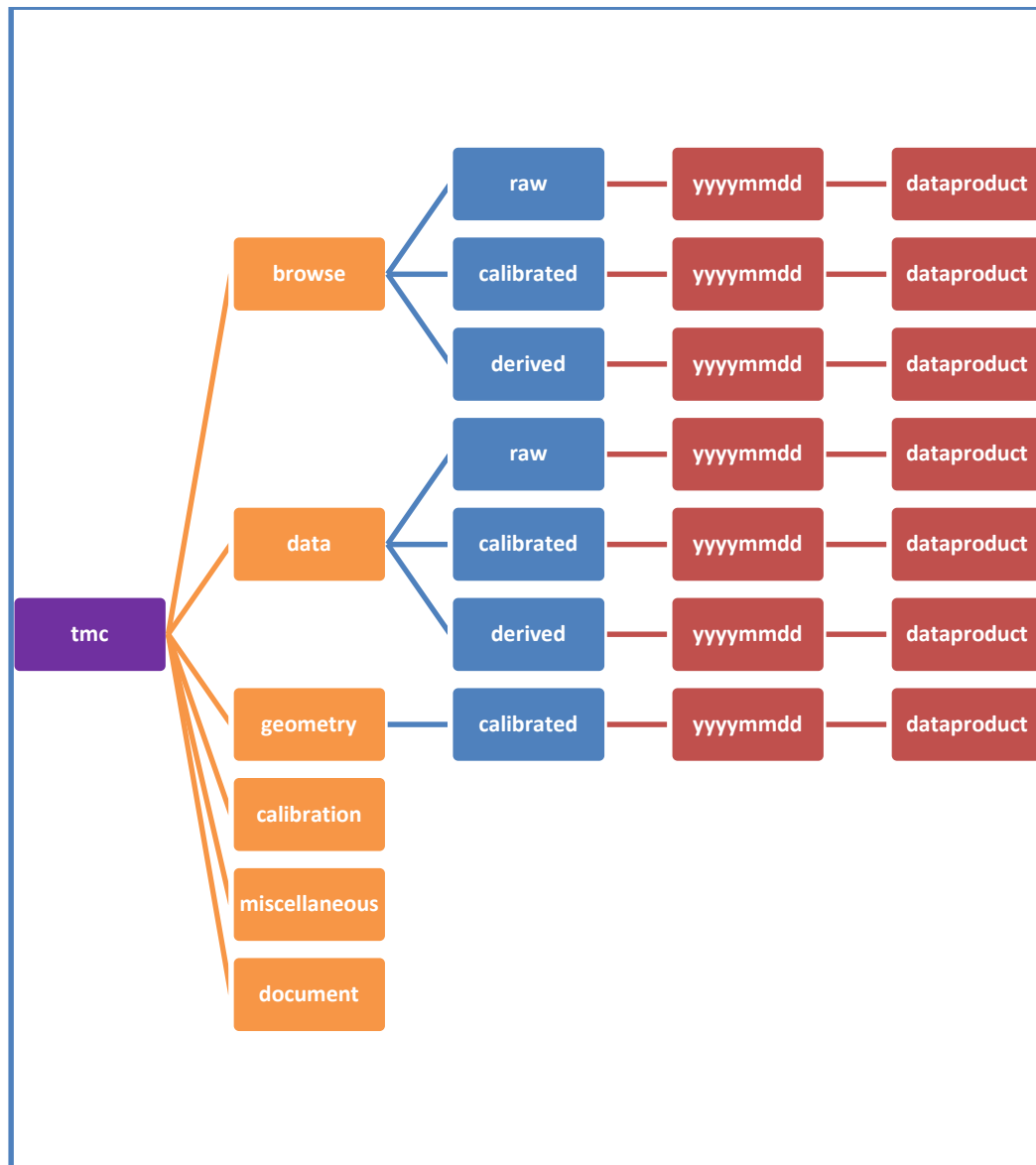


Figure 1 Instrument Collection

4. TMC2 PDS4 Data Products

The PDS4 Data Products for TMC2 is defined based on the defined data processing levels (L0, L1 and L2) as Raw, Calibrated and Derived data products. more details shown in next sub sections.

3.1 Raw Data Products

Raw Data Products contains raw data without any correction along with the system level corner coordinates. For each payload observation, PDS4 data products are generated (based on product id & sensor id derived from file name (**Annexure I**)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in **Figure 3**. For example, user has downloaded the zip product as

- ch2_tmc_nrf_20191015T102125157610840_d_img_blr.zip
- ch2_tmc_nra_20191015T102125154360840_d_img_blr.zip
- ch2_tmc_nrn_20191015T102125154360840_d_img_blr.zip

Figure 2 shows unzipped structure of TMC2 Raw Data Product zip file at user end.

```

data
├── raw
│   └── 20191015
│       ├── ch2_tmc_nra_20191015T102125154360840_d_img_blr.img
│       ├── ch2_tmc_nra_20191015T102125154360840_d_img_blr.xml
│       ├── ch2_tmc_nrf_20191015T102125157610840_d_img_blr.img
│       ├── ch2_tmc_nrf_20191015T102125157610840_d_img_blr.xml
│       ├── ch2_tmc_nrn_20191015T102125154360840_d_img_blr.img
│       └── ch2_tmc_nrn_20191015T102125154360840_d_img_blr.xml
browse
├── raw
│   └── 20191015
│       ├── ch2_tmc_nra_20191015T102125154360840_b_brw_blr.png
│       ├── ch2_tmc_nra_20191015T102125154360840_b_brw_blr.xml
│       ├── ch2_tmc_nrf_20191015T102125157610840_b_brw_blr.png
│       ├── ch2_tmc_nrf_20191015T102125157610840_b_brw_blr.xml
│       ├── ch2_tmc_nrn_20191015T102125154360840_b_brw_blr.png
│       └── ch2_tmc_nrn_20191015T102125154360840_b_brw_blr.xml
miscellaneous
├── raw
│   └── 20191015
│       ├── ch2_tmc_nrn_20191015T102125154360840_d_img_blr.lbr
│       ├── ch2_tmc_nrn_20191015T102125154360840_d_img_blr.oat
│       ├── ch2_tmc_nrn_20191015T102125154360840_d_img_blr.oath
│       └── ch2_tmc_nrn_20191015T102125154360840_d_img_blr.spm

```

Figure 2 TMC2 Raw Data Archive

Below table shows the details of individual data products (with the data formats) present inside the zip product. To visualize the data products please refer to **Annexure II and III**.

Table 4 TMC2 Raw Data Products

PDS4 Data Product Levels	PDS4 Data Products	Data Formats	Description	Visualization tools available
For sensor (TMC 2) – Fore Aft Nadir Raw	Data Raw Image	Binary	The PDS Raw data in generic binary	Any binary image viewer – imageJ, envi, erdas, snap Provide image width, image height and data type from label file
	Data Raw Label	XML	The PDS Label Product contains description about the Raw Product.	PDS4 Viewer
	Browse Raw Image	PNG	Sub sampled browse raw image	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Browse Raw Label	XML	The PDS Label Product contains description about the Browse Raw Product.	
	Miscellaneous Data (non PDS data)	Text	Contains four files Orbit & Attitude (oat) Orbit & Attitude header(oath) Liberation angle (lbr) Sun parameter file spm	Any text viewer software will be able to view the data

3.2 Calibrated Data Products

Calibrated Data Products contains radiometrically corrected data along with the refined corner coordinates. For each payload observation, PDS4 data products are generated (based on product id derived from file name (**Annexure I**)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in **Figure 4**. For example, user has to download the zip product as

- ch2_tmc_ncf_20191015T102125157610840_d_img_blr.zip
- ch2_tmc_nca_20191015T102125154360840_d_img_blr.zip
- ch2_tmc_ncn_20191015T102125154360840_d_img_blr.zip

Below figure shows unzipped structure of TMC2 Calibrated Data Product zip file at user end.

```

data
├── calibrated
│   └── 20191015
│       ├── ch2_tmc_nca_20191015T102125154360840_d_img_blr.img
│       ├── ch2_tmc_nca_20191015T102125154360840_d_img_blr.xml
│       ├── ch2_tmc_ncf_20191015T102125157610840_d_img_blr.img
│       ├── ch2_tmc_ncf_20191015T102125157610840_d_img_blr.xml
│       ├── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.img
│       └── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.xml
browse
├── calibrated
│   └── 20191015
│       ├── ch2_tmc_nca_20191015T102125154360840_b_brw_blr.png
│       ├── ch2_tmc_nca_20191015T102125154360840_b_brw_blr.xml
│       ├── ch2_tmc_ncf_20191015T102125157610840_b_brw_blr.png
│       ├── ch2_tmc_ncf_20191015T102125157610840_b_brw_blr.xml
│       ├── ch2_tmc_ncn_20191015T102125154360840_b_brw_blr.png
│       └── ch2_tmc_ncn_20191015T102125154360840_b_brw_blr.xml
miscellaneous
├── raw
│   └── 20191015
│       ├── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.lbr
│       ├── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.oat
│       ├── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.oath
│       └── ch2_tmc_ncn_20191015T102125154360840_d_img_blr.spm

```

Figure 3 TMC2 Calibrated Data Archive

Table 5 shows the details of individual data products (with the data formats) present inside the zip product. To visualize the data products please refer to **Annexure II** and **III**

Table 5 TMC2 Calibrated Data Products

PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Description	Visualization tools available
For sensor – Fore Aft Nadir Calibrated	Data Calibrated Image	Binary	The PDS Radio metrically corrected data in generic binary format	Any binary image viewer – imageJ, envi, erdas, snap Provide image width, image height and data type from label file
	Data Calibrated Label	XML	The PDS Label Product contains description about the Calibrated Product.	PDS4 Viewer
	Browse Calibrated Image	PNG	Sub sampled browse calibrated image	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Browse Calibrated Label	XML	The PDS Label Product contains description about the Browse Calibrated Product.	
	Geometric Calibrated Grid	CSV	A grid file containing lon, lat, scan and pix as four columns along with the data	Any text editor, ms excel can be used to load the csv file.
	Geometric Calibrated Grid Label	XML	The PDS Label Product contains description about the Geometry Calibrated Product.	PDS4 Viewer
	Miscellaneous Data (non PDS data)	Text	Contains four files Orbit & Attitude (oat) Orbit & Attitude header(oath) Liberation angle (lbr) Sun parameter file spm	Any text viewer software will be able to view the data

3.3 Derived Data Products

Derived Data Products contains radiometrically and geometrically corrected data along with the corner coordinates. For each payload observation, PDS4 data products are generated (based on product id derived from file name (Annexure I)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in figure2.

For example, user has to download the zip product as

- ch2_tmc_ndn_20191015T102125154360840_d_dtm_blr.zip
- ch2_tmc_ndn_20191015T102125154360840_d_oth_blr.zip

Figure 5 below shows unzipped structure of TMC2 Calibrated Data Product zip file at user end.

```

data
├── derived
│   └── 20191015
│       ├── ch2_tmc_ndn_20191015T102125154360840_d_dtm_blr.tif
│       ├── ch2_tmc_ndn_20191015T102125154360840_d_dtm_blr.xml
│       ├── ch2_tmc_ndn_20191015T102125154360840_d_oth_blr.tif
│       └── ch2_tmc_ndn_20191015T102125154360840_d_oth_blr.xml
browse
├── derived
│   └── 20191015
│       ├── ch2_tmc_ndn_20191015T102125154360840_b_bdt_blr.png
│       ├── ch2_tmc_ndn_20191015T102125154360840_b_bdt_blr.xml
│       ├── ch2_tmc_ndn_20191015T102125154360840_b_bot_blr.png
│       └── ch2_tmc_ndn_20191015T102125154360840_b_bot_blr.xml

```

Figure 4 TMC2 Derived Data Archive

Table 6 shows the details of individual data products (with the data formats) present inside the zip product.

Table 6 TMC2 Derived Data Products

PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Description	Visualization tools available
Derived	Data DEM Image	Geo Tiff	DEM image data	Any Tiff image viewer – imageJ, envi, erdas
	Data DEM Label	XML	The PDS Label Product contains description about the Derived DEM data Product.	
	Data Ortho Image	Geo Tiff	Ortho rectified image data	Any Tiff image viewer – imageJ, envi, erdas
	Data Ortho Label	XML	The PDS Label Product contains description about the Derived Ortho Product.	
	Browse DEM Image	PNG	Sub sampled browse DEM image	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Browse DEM Label	XML	The PDS Label Product contains description about the Browse Derived DEM data Product.	
	Browse Ortho Image	PNG	Sub sampled browse ortho image	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Browse Ortho Label	XML	The PDS Label Product contains description about the Browse Derived Ortho Product.	

5. Archive Products formats

Data that comprises the TMC 2 archives are formatted in accordance with PDS specifications. This section provides details on the formats used for each of the products included in the archive. The detailed file formats for the defined data products based on PDS4 data types have been defined shown in next sub section.

5.1 Data File Formats

TMC 2 instrument is an imaging payload. The primary data collected from instrument is raw image. The TMC 2 imager has three sensors – Fore, Aft and Nadir. For each sensor both primary as well as secondary data sets are defined that contains data product levels, types and formats shown in the table 7.

Table 7 TMC2 PDS4 Data Products

Sensors	PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Data type*
For each sensor - Fore, Aft and Nadir	Raw	Image	Binary	UnsignedLSB2
	Raw	Browse	PNG	UnsignedByte
	Calibrated	Image	Binary	UnsignedLSB2
	Calibrated	Browse	PNG	UnsignedByte
	Calibrated	Geometry	CSV	ASCII Text
Stereo / Triplet Combination	Derived	DEM	Geo Tiff	SignedLSB2 (Float/ double)
	Derived	Ortho	Geo Tiff	UnsignedLSB2
	Derived	Browse DEM	PNG	UnsignedByte
	Derived	Browse Ortho	PNG	UnsignedByte

Note*:

UnsignedLSB2 is UInt16 (Unsigned Integer of 16-bit length)

SignedLSB2 is Int16 (Signed Integer of 16-bit length)

6. Local Data Dictionary

Every mission has some conditions based on the events that needs to be handle as an information at output product level. Some information is common to every planetary mission launched by ISRO. There is an ISRO Science Data Archive established during Chandrayaan-1 mission in 2008. All the products hosted for users under ISDA are PDS compliance. For Chandrayaan-2 mission, PDS4 Archive standard was adopted. PDS4 is currently next and future generation archive standard adopted by all the space agencies including NASA, ESA and JAXA for their current and future mission. To handle the ISRO planetary mission's specific parameters, there is a concept in PDS4 called Local Data Dictionary i.e. LDD. In LDD one can define all the parameters and that can be used as a reference for future planetary missions in India. LDD takes basic data dictionary of PDS4 and then top of it builds local dictionary. For Chandrayaan-2 mission and future planetary missions of ISRO, list of parameters is identified shown in details in below table.

Table 8 ISDA Local Data Dictionary Class and Attributes List

Class/Sub Class Hierarchy	Attribute Name	Description/Value
Mission_Area Product_Parameters	job_id	Unique id assign by DP Scheduler when the product is ingested
	level0_dir_name	DP Input Level-0 directory name
	imaging_orbit_number	Imaging Orbit Number assigned on board
	dumping_orbit_number	Dumping Orbit Number is assigned based on the data collected on ground station
	line_exposure_duration	Image integration time
	gain	Gain is one of the operating modes. It is value is set as g1
	exposure	exposure is one of the operating modes. It is value is set as e1 or e2
	detector_pixel_width	Pixel width of detector
	focal_length	Instrument focal length
	spacecraft_altitude	Spacecraft height
	orbit_limb_direction	Orbit limb direction it can have two values - Ascending or Descending
	spacecraft_yaw_direction	

		Spacecraft yaw mode it can have two values – True/False a) True means Reverse, in flight yaw mode Reverse describes 180 degree rotated from nominal. b) False means Forward, in flight yaw mode Forward describes 0 degree rotated from nominal) There are four possible orbit_limb_direction and spacecraft_yaw_direction combinations: 1. Descending and False, no changes in lines or samples are performed 2. Descending and True, samples are reversed 3. Ascending and False, both lines and samples are reversed 4. Ascending and True. Lines are reversed
	reference_data_used	Reference data set used in refining the system level corners
	pixel_resolution	Image pixel resolution expressed in unit m/pixel
Mission_Area • Geometry_Parameters ○ System_Level_Coordinates ○ Refined_Level_Coordinates ○ Corrected_Level_Coordinates	upper_left_latitude	Upper left latitude reticle point with number 1
	upper_left_longitude	Upper left longitude reticle point with number 1
	upper_right_latitude	Upper right latitude reticle point with number 3
	upper_right_longitude	Upper right longitude reticle point with number 3
	lower_left_latitude	Lower left latitude reticle point with number 7
	lower_left_longitude	Lower left longitude reticle point with number 7
	lower_right_latitude	Lower right latitude reticle point with number 9
	lower_right_longitude	Lower right longitude reticle point with number 9

7. Conclusions

This user guide is abstract view of PDS4 Data Products. As per the mission planning, normal operations plan (nop) was begin in November 2019. Data was collected from the onboard payload at ground station for further processing and PDS4 data products were generated and archived under active archive structure. After defined lock in period as per SSPO ISRO HQ, Long Term Archive (LTA) was prepared and interfaced with Chandrayaan-2 Map based browse application hosted from ISSDC for the users. LTA is self-structured, self-documented and peer reviewed data sets. For user's assistance contact information about the team is added under **Annexure III**.

Annexure – I

File Naming Conventions & Formats

This Annexure describes the naming convention used for science data files for the Chandrayaan-2 mission. Table 9 shows the detail description about the following file naming conventions designed for the data products.

ch2_<inst>_<mtc>_<YYYYMMDDTHHMMSSsss>_<p>_<prd>_<stn>.fff

Table 9 File Naming Conventions

Code	Description
ch2	It denotes the Mission name whose value is fixed as ch2
inst	It denotes the Instrument id, it can have following value = tmc (refers to the TMC2)
m	It denotes the mission phase name, it can have following value = n (refers to the Normal Operations Phase - nop)
t	It denotes the data type; it can have following values: = r (it denotes raw data) = c (It denotes calibrated data) = d (It denotes derived data)
c	It denotes the imaging mode/camera id/band. It can be one of the following: For tmc2 = f (It denotes the fore camera) = a (It denotes the aft camera) = n (it denotes the nadir camera)

YYYYMMDDTHHMMSSsss	It denotes the observation/imaging start time in format [YYYYMMDDTHHMMSSsss] YYYY = stand for year MM = stand for month in MM format DD = stand for day in DD format T = stand for time. HH = stand for hour in HH format MM = stand for minute in MM format SSS = stand for seconds in SS format sss = stand for 1/10 millisecond or 100 microsecond in ssss format
p	It denotes the PDS Data Products categories. It value can be any of the following: = d (It denotes data products under data directory) = b (it denotes browse data products under browse directory) = g (it denotes gridded data products under geometry directory)
prd	It denotes the PDS Data Product Name. It value can be any of the following: = img (It denotes the image data product) = brw (It denotes the browse data product) = grd (It denotes the gridded data product) = dtm (it denotes DTM derived product) = oth (it denotes Ortho derived product)
stn	It denotes the station id from which data was acquired. It value can be any of the following: = d32 (ISSDC Bangalore) = d18 (ISSDC Bangalore) = gds (Gold Stone USA) = cnb (Canberra, Australia)

fff	It denotes the standard file type extension. It can have any one of the following values: = img (PDS Image Data File) = xml (PDS Detached Label File) = jpg (PDS Browse Data File) = csv (PDS Geometry Data File) = tif (PDS Geo tiff data for derived products)
-----	---

Annexure – II

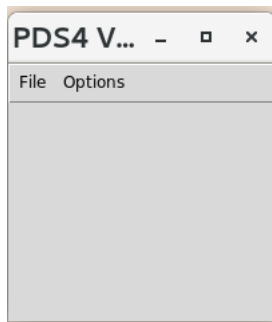
Visualize PDS4 Data Products using PDS4 Viewer

There are many PDS tools available for opening the Raw/Calibrated Data products.

PDS4Viewer - PDS4Viewer is a simple display tool for visualizing data contained in the PDS4 data standard. PDS4Viewer tool can be downloaded from internet using link “http://sbndev.astro.umd.edu/wiki/PDS4_Viewer”. This Software is available for both Linux as well as window platform. The given data sets can be opened using PDS4Viewer.

The instructions to open the data using PDS4Viewer are as follows:

1. Download the PDS4Viewer tool from internet http://sbndev.astro.umd.edu/wiki/PDS4_Viewer. There will be a zip file (for both Linux/Windows) contains the PDS4Viewer.
2. Extract the Zip file into any location.
3. Go to extract the directory, run pds4_viewer. The PDS4Viewer GUI will come which is shown below.



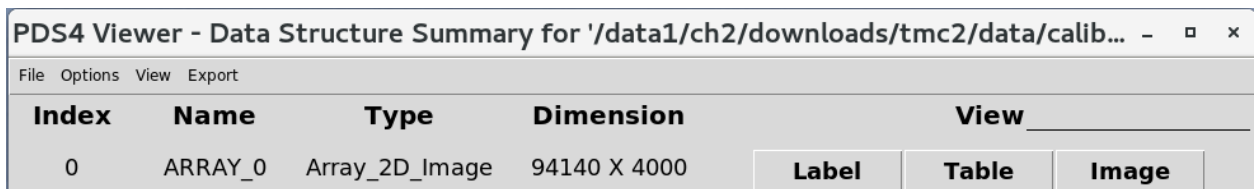
Opening PDS4 Raw/Calibrated Image Product

=====

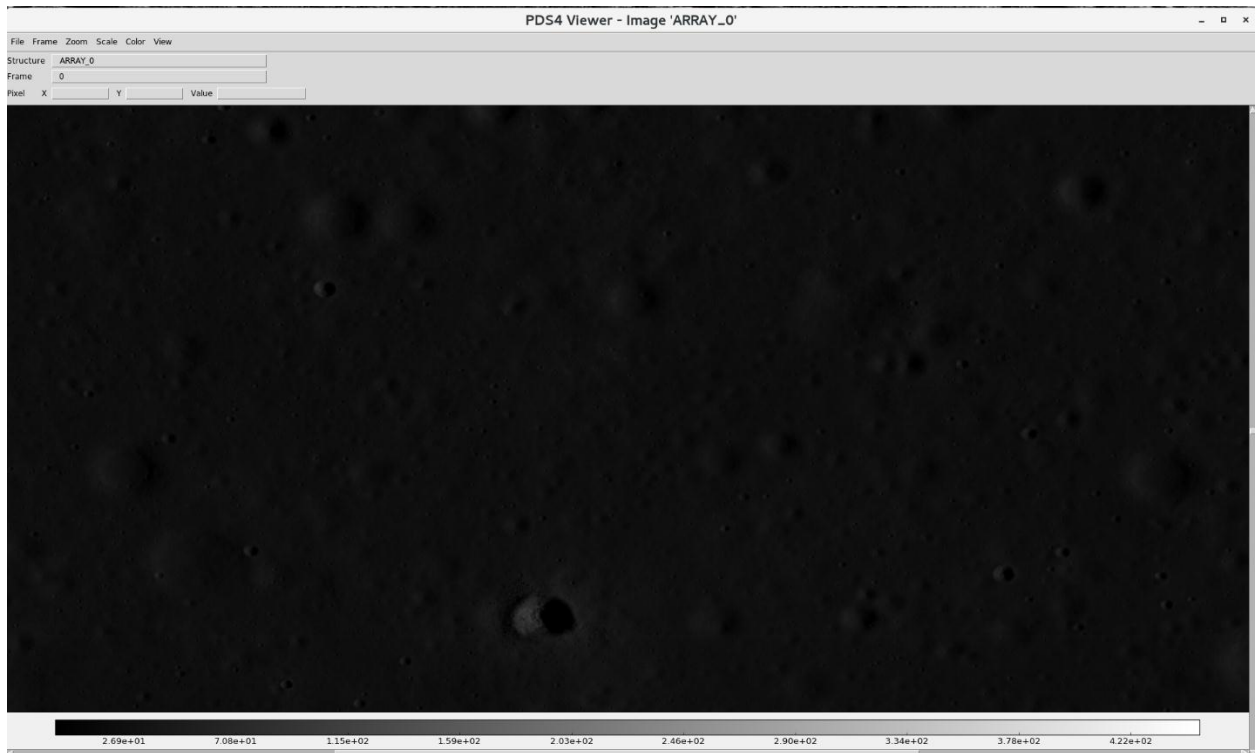
1. After opening the PDS4Viewer display tool. Click on the File menu and select Open Object
2. Click file open dialog will open which is shown below select Label File and click Open button



3. After clicking Open button, the data will be loaded in the PDS4Viewer shown below:



4. For Opening an image, click on the Image button shown on the above figure at step no. 6. Below shows the image from the TMC2 Payloads.



5. For Opening a label, click on the Label by selecting from View drop down menu. Below figures shows the label file as label view and xml view

PDS4 Viewer - Label View

File Edit View

Label

Product_Observational

Identification_Area

logical_identifier: urn:isro:isda:ch2_cho:tmc_calibrated:ch2_tmc_ncn_20191015t102125154360840_d_img_blr
version_id: 1.0
title: Chandrayaan-2 Orbiter TMC2 Experiment
information_model_version: 1.11.0.0
product_class: Product_Observational

Modification_History

Modification_Detail

modification_date: 2019-05-07
version_id: 1.0
description: PDS4 product label created by SAC Optical DP Team

Observation_Area

Time_Coordinates

start_date_time: 2019-10-15T10:21:25.154360840Z
stop_date_time: 2019-10-15T10:26:29.790141600Z

Primary_Result_Summary

purpose: Science
processing_level: Calibrated

Investigation_Area

name: chandrayaan-2
type: Mission

Internal_Reference

lidvid_reference: urn:isro:isda:context:investigation:mission.chandrayaan2::1.0
reference_type: data_to_investigation

Observing_System

Observing_System_Component

name: Chandrayaan 2 Orbiter
type: Spacecraft
description:
Chandrayaan-2 Orbiter is an Orbiter craft under the Chandrayaan-2 Spacecraft consists of various scientific instruments.

Observing_System_Component

name: terrain mapping camera
type: Instrument
description:
TMC-2 (Terrain Mapping Camera) is one of the scientific instrument hosted on the Chandrayaan-2 Orbiter. The instrument is passive electro-optical imaging camera operating in visible-panchromatic band (500-800nm), comprising three optical sensors Fore, Nadir and Aft placed with angles of -25 deg, 0 deg and +25 deg to have three stereo views of the same target as spacecraft moves along the track.

Target_Identification

name: Moon
type: Satellite
description: Moon is a natural satellite of Earth

Mission_Area

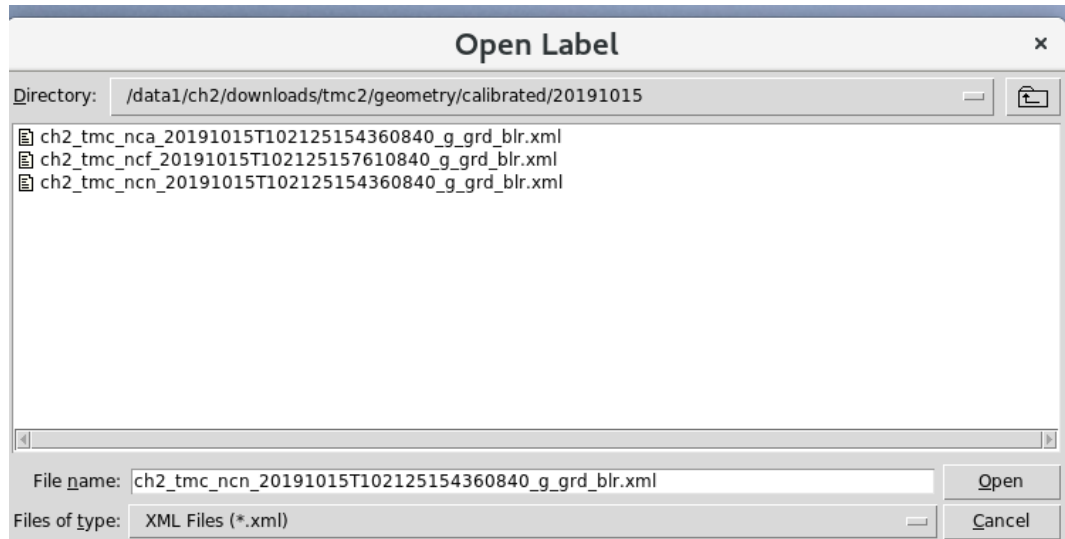
Product_Parameters

imaging_orbit_number: 624
dumping_orbit_number: 628
image_observation_type: regular
instrument_name: TMC2

Search ☐ Match Case

Opening PDS4 Geometry Grid Data Product

1. After opening the PDS4Viewer display tool. Click on the File menu and select Open Object
2. Click file open dialog will open which is shown below select Label File and click Open button



3. After clicking Open button, the data will be loaded in the PDS4Viewer shown below:

PDS4 Viewer - Data Structure Summary for '/data1/ch2/downloads/tmc2/geometry/calibrat...					
File Options View Export					
Index	Name	Type	Dimension	View	
1	STREAM_CSV_ID	Table_Delimited	4 cols X 38663 rows	Label	Table Plot

4. For Opening grid file, click on the table button shown on the above figure at step no. 6. Below shows the grid file from the TMC2 Payloads

PDS4 Viewer - Table 'STREAM_CSV_ID'

File Options View

Row #	Longitude	Latitude	Pixel	Scan
0	0.6147276	-7.70536646	0	0
1	53.6014937	-7.70516626	100	0
2	53.588262	-7.70496572	200	0
3	53.5750325	-7.70476485	300	0
4	53.5618051	-7.70456365	400	0
5	53.5485797	-7.70436212	500	0
6	53.5353564	-7.70416026	600	0
7	53.5221351	-7.70395806	700	0
8	53.5089157	-7.70375553	800	0
9	53.4956981	-7.70355267	900	0
10	53.4824824	-7.70334947	1000	0
11	53.4692685	-7.70314594	1100	0
12	53.4560564	-7.70294208	1200	0
13	53.4428459	-7.70273787	1300	0
14	53.4296371	-7.70253333	1400	0
15	53.4164299	-7.70232846	1500	0
16	53.4032243	-7.70212324	1600	0
17	53.3900202	-7.70191769	1700	0
18	53.3768175	-7.7017118	1800	0
19	53.3636163	-7.70150557	1900	0
20	53.3504164	-7.701299	2000	0
21	53.3372179	-7.70109208	2100	0
22	53.3240206	-7.70088483	2200	0
23	53.3108246	-7.70067724	2300	0
24	53.2976298	-7.7004693	2400	0
25	53.2844361	-7.70026102	2500	0
26	53.2712436	-7.70005239	2600	0
27	53.258052	-7.69984342	2700	0
28	53.2448615	-7.69963411	2800	0
29	53.2316719	-7.69942445	2900	0
30	53.2184833	-7.69921444	3000	0
31	53.2052955	-7.69900409	3100	0
32	53.1921086	-7.69879339	3200	0
33	53.1789224	-7.69858234	3300	0
34	53.1657369	-7.69837095	3400	0
35	53.1525522	-7.6981592	3500	0
36	53.1393681	-7.6979471	3600	0
37	53.1261845	-7.69773465	3700	0
38	53.1130016	-7.69752185	3800	0
39	53.0998191	-7.6973087	3900	0
40	53.0867689	-7.69709733	3999	0
41	53.6149405	-7.68847266	0	100
42	53.6017064	-7.68827248	100	100

5. For Opening a label, click on the Label by selecting from View drop down menu. Below figures shows the label file as label view and xml view

PDS4 Viewer - Label View

File Edit View

Label

Product_Observational

Identification_Area
logical_identifier: urn:isro:isda:ch2_cho:tmc_calibrated:ch2_tmc_ncn_20191015t102125154360840_g_grd_blr
version_id: 1.0
title: Chandrayaan-2 Orbiter TMC 2 Experiment
information_model_version: 1.11.0.0
product_class: Product_Observational

Modification_History

Modification_Detail
modification_date: 2019-05-07
version_id: 1.0
description: PDS4 product label created by SAC Optical DP Team

Observation_Area

Time_Coordinates
start_date_time: 2019-10-15T10:21:25.154360840Z
stop_date_time: 2019-10-15T10:26:29.790141600Z

Investigation_Area
name: Chandrayaan-2
type: Mission

Internal_Reference
lvidid_reference: urn:isro:isda:context:investigation:mission.chandrayaan2::1.0
reference_type: data_to_investigation

Observing_System

Observing_System_Component
name: Chandrayaan 2 Orbiter
type: Spacecraft
description:
Chandrayaan-2 Orbiter is an Orbiter craft under the Chandrayaan-2 Spacecraft consists of various scientific instruments.

Observing_System_Component
name: terrain mapping camera
type: Instrument
description:
TMC-2 (Terrain Mapping Camera) is one of the scientific instrument hosted on the Chandrayaan-2 Orbiter. The instrument is passive electro-optical imaging camera operating in visible-panchromatic band (500-800nm), comprising three optical sensors Fore, Nadir and Aft placed with angles of -25 deg, 0 deg and +25 deg to have three stereo views of the same target as spacecraft moves along the track.

Target_Identification
name: Moon
type: Satellite
description: Moon is a natural satellite of Earth

File_Area_Observational

File
file_name: ch2_tmc_ncn_20191015T102125154360840_g_grd_blr.csv
local_identifier: STREAM DELIMITED CSV FILE ID
creation_date_time: 2019-10-29T15:36:39.039
file_size: 1314312
records: 38663
md5_checksum: daa12eaaa9d189ace9f4cbee6a6cc1ec
comment:
This File contains selenographic coordinates for the image points.
md5_checksum is provided for ensuring data integrity when users are downloading the data.

Header
name: Column headings for Grid Data
local_identifier: HEADER
offset: 0
object_length: 31
parsing_standard_id: PDS DSV 1

Table_Delimited
local_identifier: STREAM_CSV_ID
offset: 33
parsing_standard_id: PDS DSV 1
records: 38663
record_delimiter: Carriage-Return Line-Feed
field_delimiter: Comma

Record_Delimited
fields: 4

Annexure – III

Visualize PDS4 Data Products using SNAP

SNAP – Sentinel Application Platform (SNAP) is very fast image display and navigation tool developed by ESA for visualizing data products. SNAP tool can be downloaded from internet using link “<http://step.esa.int/main/download/snap-download>”. This Software is available for both Linux as well as window platform. The given data sets can be opened using SNAP. Following steps are required for opening the data products using SNAP tool.

- Go to the PDS archive area and the concerned folder for the data product.

```
[ch2opr@raman 20200708]$ ls -ltrh
total 6.0G
-rw-rw-rw- 1 root bin 1.5G Sep 30 17:40 ch2_tmc_ncf_20200708T1922155963_d_img_d18.img
-rw-rw-rw- 1 root bin 1.5G Sep 30 17:41 ch2_tmc_nca_20200708T1922156288_d_img_d18.img
-rw-rw-rw- 1 root bin 1.5G Sep 30 17:42 ch2_tmc_ncn_20200708T1922155963_d_img_d18.img
-rw-rw-rw- 1 root bin 7.4K Oct 1 11:25 ch2_tmc_ncf_20200708T1922155963_d_img_d18.xml
-rw-rw-rw- 1 root bin 7.4K Oct 1 11:26 ch2_tmc_nca_20200708T1922156288_d_img_d18.xml
-rw-rw-rw- 1 root bin 7.4K Oct 1 11:26 ch2_tmc_ncn_20200708T1922155963_d_img_d18.xml
-rw-rw-rw- 1 root bin 556M Oct 1 11:29 ch2_tmc_ncf_20200708T1922155963_d_img_d18.zip
-rw-rw-rw- 1 root bin 497M Oct 1 11:32 ch2_tmc_nca_20200708T1922156288_d_img_d18.zip
-rw-rw-rw- 1 root bin 528M Oct 1 11:35 ch2_tmc_ncn_20200708T1922155963_d_img_d18.zip
[ch2opr@raman 20200708]$ pwd
/lunas/projects/ch2/orbiter/lunar_bound_phase_data/opr/workflow/output/lta/PRE_OD/isda_archive/ch2_bundle/cho_bundle/nop/tmc_collection/data/calibrated/20200708
[ch2opr@raman 20200708]$
```

- Image data is in *img form. To get the dimensions of the image file, execute ‘cat’ command on the concerned *xml file.

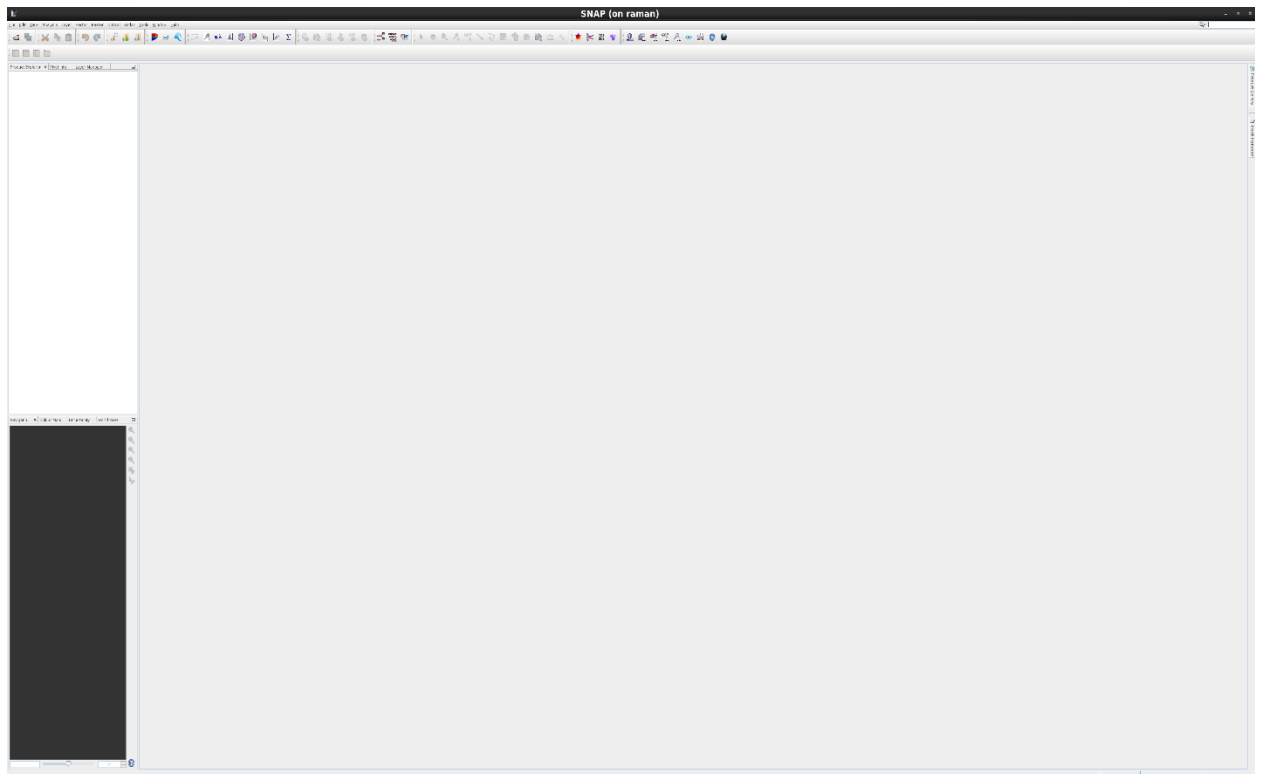
```
</File>
<Array_2D_Image>
  <offset unit="byte">0</offset>
  <axes>2</axes>
  <axis_index_order>Last Index Fastest</axis_index_order>
  <Element_Array>
    <data_type>UnsignedLSB2</data_type>
  </Element_Array>
  <Axis_Array>
    <axis_name>Line</axis_name>
    <elements>197812</elements>
    <sequence_number>1</sequence_number>
  </Axis_Array>
  <Axis_Array>
    <axis_name>Sample</axis_name>
    <elements>4000</elements>
    <sequence_number>2</sequence_number>
  </Axis_Array>
</Array_2D_Image>
</File_Area_Observational>
```

The last section of the file contains axis_name ‘Line’ and axis_name ‘Sample’. They represent scanlines and pixels of the image. ‘Elements’ tag contain corresponding values. Datatype is present in ‘data_type’ tag. Here it is unsigned int16.

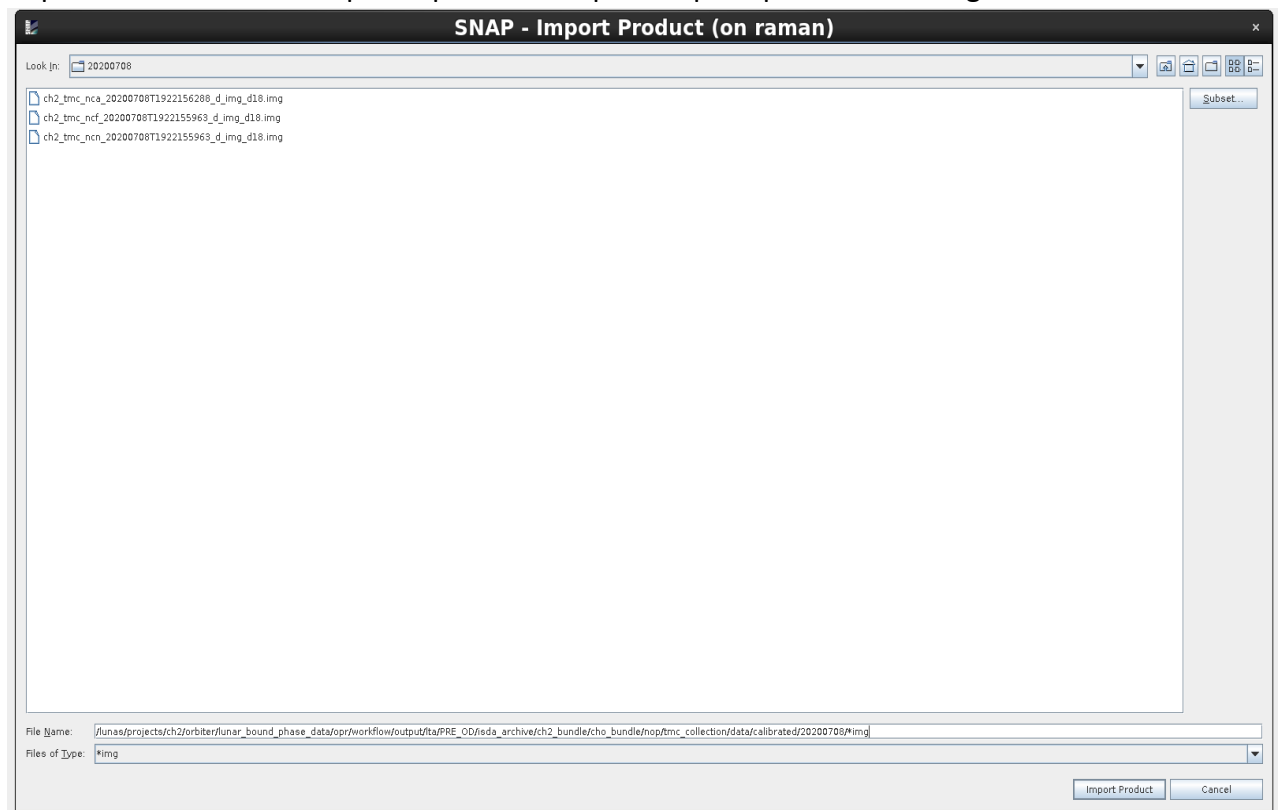
- Type snap command on command line and keep it in background.

```
[ch2opr@raman 20200708]$ snap &
[1] 26882
```

- SNAP window opens up.



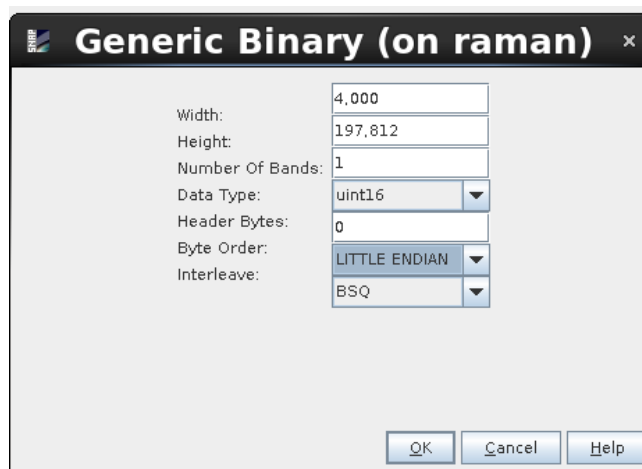
- Select File>Import>SAR Formats>Generic Binary.
- Import Product Window opens up. Enter the path of pds4 product list *.img data.



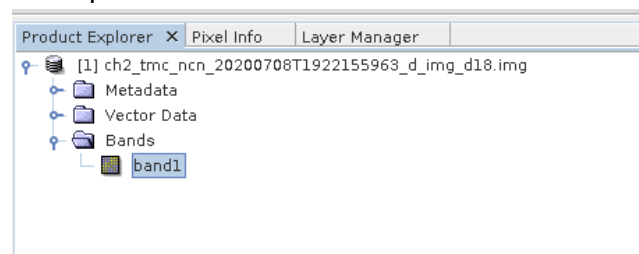
- Select required file *.img file.



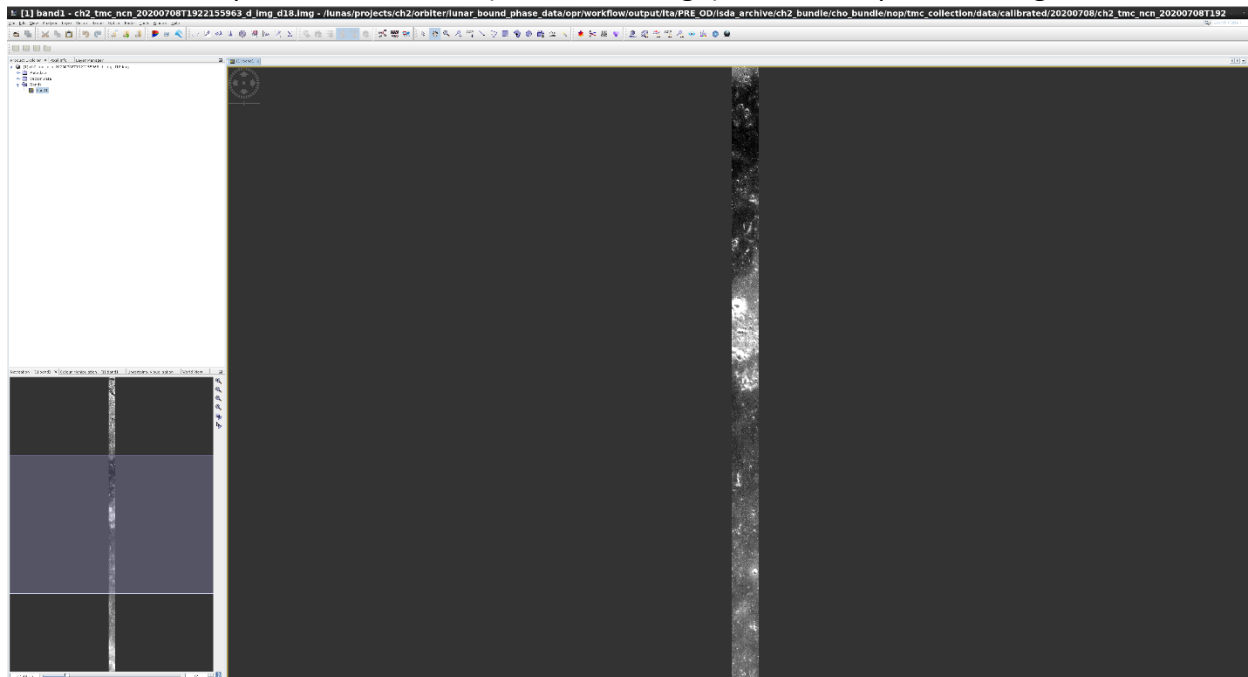
- Enter the required dimensions and parameters as seen in the *.xml file.



- Product Explorer lists the product. Select band1.



- Select the hand option in the toolbar(above the image) to zoom or pan the image.



Annexure – IV**Team: Contact Information**

Teams	Person Name	Contact	Email Id
Operations Team @ ISSDC IDSN Bylalu, Bengaluru	Sri. Sreenath	080-2809-4416/17/18 (SCC), 2809-4419 (MOX), 2202-9173/74 (ISSDC-IDSN)	sreenath@istrac.gov.in
Principal Investigator (PI) Team @ SAC Ahmedabad	DR. A S Arya	07926914107	arya_as@sac.isro.gov.in